



Timing: 2:00 to 5:00 PM

IV Semester

Max mark: 50

All questions are Compulsory

1. Consider a library management system with the following key entities:

- Library manages multiple Books and Members.
- A Book can be borrowed by a Member
- Employee is a type of Member and is responsible for managing the library's operations.
- A Member can borrow multiple Books
- Library depends on Employee for operational tasks
- Member and Employee share an inheritance relationship, with Employee being a specialized type of Member.

(a) Write the code for the classes Library, Book, Member, and Employee in any object-oriented language. Implement the relationships, including:

- Aggregation between Member and Book.
- Generalization between Member and Employee.
- Dependency between Library and Employee.

(5)

(b) Draw a UML class diagram for the system, clearly labeling the relationships:

- Association between Library and Book.
- Aggregation between Member and Book.
- Generalization between Member and Employee.
- Dependency between Library and Employee.

(5)

2. Given the following system with three modules: Module A, Module B, and Module C, where the arrows represent dependency relationships between the modules:

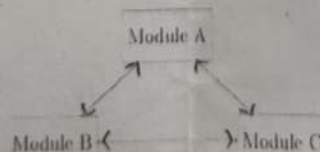


Figure 1: Showing Module Dependencies.

(a) Calculate the Instability Index (I) and Abstractness (A) for each module.

(5)

- (b) Analyze the system's modularity and stability based on your calculations. Discuss the potential risks if the instability index is not optimized and suggest strategies to improve the system's maintainability.

(5)

3. (a) Explain the core concepts of Git, including repositories, commits, branches, and merging.

(5)

- (b) Consider a scenario where two developers, Developer A and Developer B, are working on the same file in different branches. Developer A commits changes to feature-branch, while Developer B makes modifications to the same file in main. After Developer A attempts to merge feature-branch into main, a merge conflict occurs.

- Explain in detail the steps to resolve this merge conflict using Git.
- Discuss how Git ensures data integrity and prevents data loss when resolving conflicts.

(5)

4. You are managing a complex software project with the following characteristics: Scope: Large system with functional and non-functional requirements, Timeline: Strict deadline of 6 months, Client Expectations: Clear initial requirements, but some flexibility for changes, Risk Level: High due to complexity and the possibility of changing requirements during development. The table below compares three software development models (A, B, and C) with key metrics:

Model	Time (months)	Dev. Cost (\$)	Adapt Cost (\$)
A	6	150,000	40,000
B	5	140,000	30,000
C	6	160,000	20,000

*Note: "Adapt Cost" refers to the "Cost to Adapt to Changes".

Choose the most suitable model based on the total costs, justifying your choice in terms of cost, time, and flexibility for managing changes. Also, evaluate the nature of the model (predictive vs. adaptive) and explain how each model fits the project's requirements.

(10)

5. You are designing a Data Flow Diagram (DFD) for a Library Management System that handles Book Checkout, Book Return, and Inventory Update processes. The system checks the availability of books, issues books, returns books, calculates overdue fines, and allows the librarian to update book details. Draw a Level 0 and Level 1 DFD for the system, explain the anomalies of Miracle, Black Hole, and Grey Hole with examples, and identify which anomaly is present if the system fails to calculate overdue fines, suggesting a solution.

(10)

*****Ends Here*****

A lagega
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Email: sonapatiit@gmail.com, website: www.iiitsonepat.ac.in

B.Tech.2nd End Semester Examination, 19th May 2025.

Subject: Operating System

Subject Code: UCS013

Roll No.

1 2 3 1 2 0 2 1

Branch: IT

Max.Marks.-50 (6+6+5+15+18)

Time- 3 Hours

1. Let $P_1, P_2, \dots, P_{20}$ be a set of 20 cooperating processes, all of which have access to a shared integer X . For $i \in \{1, 2, 3, \dots, 20\}$, process P_i calls `do_something(i)`, the code for which is given below.

```
shared int X = 0;
void do_something (int i)
{ X += i; printf("%d %d ", i, X); }
```

(a) What is the minimum possible value of X after all 20 processes complete execution? Justify the answer.

(b) Your task is to synchronize the execution of the above processes using semaphores so that the combined output of all 20 processes is as follows.
11 23 36... 20 210.

- (i) Show how you would declare and initialize the necessary semaphore(s). The following examples show how semaphores can be declared and initialized.
shared semaphore $S = 1$;
shared semaphore $\text{mutex}[m] = \{0, 0, \dots, 0\}$;
- (ii) Modify the body of the function `do_something(i)` by using appropriate `wait()` and `signal()` calls to synchronize $P_1, P_2, \dots, P_{20}$ so that the messages are printed in the aforesaid order.

2. A computer system uses the Banker's Algorithm to deal with deadlocks. Its current state is shown in the tables below, where P_0, P_1, P_2 are processes and R_0, R_1, R_2 are resource types.

Max Need				Current Allocation				Available		
	R_0	R_1	R_2		R_0	R_1	R_2	R_0	R_1	R_2
P_0	4	1	2	P_0	1	0	2	2	2	0
P_1	1	5	1	P_1	0	3	1			
P_2	1	2	3	P_2	1	0	2			

- (a) Show that the system can be in this state.
 (b) What will the system do on a request by process P0 for one unit of resource type R1?
3. Consider a system with byte-addressable memory, 32 bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. What is the size of the page table in the system in megabytes?
4. a) Apply LRU and Optimal page replacement policies for the following reference string and calculate how many page-faults would occur for three page frame structure:
 1, 2, 3, 4, 5, 5, 3, 4, 9, 6, 2, 7, 8, 7, 3, 5, 9, 7, 6, 4, 5, 4, 2.
- b) Find the following statement is True/False with short explanation (i-iv):
- Under preemptive scheduling, when a process switches from the running to the ready state, it may lose control of the CPU..
 - Memory mapping a file using the mmap system call adds one or more entries to the page table of the process.
 - The read system call to fetch data from a file always blocks the invoking process. (the data may be readily available in the disk buffer cache)
 - Reading files via memory mapping them avoids an extra copy of file data from kernel space buffers to user space buffers.
 - Consider the process of opening a new file that does not exist (obviously, creating it during opening), via the "open" system call.
 Describe changes to all the in-memory and disk-based file system structures (e.g., file tables, inodes, and directories) that occur as part of this system call implementation. Write clearly, listing the structure that is changed, and the change made to it.
5. Describe the following in brief with a suitable example (working flowchart if required):
- Paging and Demand Paging
 - Process management and Memory Management
 - Segmentation with Paging
 - Dining Philosopher Problem
 - Bankers Algorithm
 - LRU and optimal page replacement algorithm

$$\begin{aligned}
 & \frac{4 \text{ byte}}{4 \times 1024 \text{ byte}} \\
 & \frac{1}{1024} \times 4 \\
 & \frac{32 \text{ B}}{4 \text{ KB}} \\
 & \frac{4 \text{ byte}}{4 \times 1024} \\
 & \text{Pages} = \frac{1}{1024}
 \end{aligned}$$



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Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in
B.Tech. End Semester Examination, May '2024.

Subject: Computer Networks

Subject Code: CSC404

Roll No.

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Branch: CSE

Max.Marks.-50

Time- 3 Hours

Note: All questions are compulsory and your answer should include a rough sketch, if necessary.

- Q-1 a) Define computer networks. (1+2+2)=5 Marks
b) Define the fundamental characterise of the effective data communication ?
c) Overview of different transmission mode ?
- Q-2 a) Describe the TCP/IP layer's function using suitable sketch diagram. (2.5+1.5+1)=5 Marks
b) Difference between guided media and unguided media.
c) Define Modulation.
- Q-3 a) Define P-persistent, 1-persistent, non-persistent and 0-persistent. (3+1+1)5 Marks
b) Explain MAC Address?
c) Describe main principal of Selective Repeat ARQ protocol.
- Q-4 a) Explain *Cyclic Redundancy Check* (CRC)? Suppose we want to transmit the message 11001001 and protect it from errors using the CRC polynomial $x^3 + 1$. Use polynomial long division to determine the message that should be transmitted. If the receiver receives the message, check if the receiver receives the correct message or not. (3+2)=5 Marks
b) Describe main idea of *IEEE 802.5* standard.
- Q-5 a) An ISP is granted a block of addresses for **IIITSonepat** starting with 190.100.0.0/16. The ISP needs to distribute these addresses to three groups (Academic, Exam controller and Director office) of customers as follows: (3+2)=5 Marks
1. The Academic has 64 users; each need 256 addresses.
2. The Exam Controller has 128 users; each need 128 addresses.

3. The Director office has 128 users; each need 64 addresses.

Design the subblocks and find out how many addresses are still available in **IIIT Sonepat** after these allocations.

- b) Discuss *Distance Vector Routing Protocol* ?

- Q-6 a) A block of addresses is granted to a small organization. We know that one of the addresses is *205.16.37.39/28*. What is the first address and total number of address in the block? (2+1+2)=5 Marks

- b) Define working principal of NAT.
c) Sketch the format of a *IPv6 datagram*.

- Q-7 a) Leaky bucket is the well know traffic shaping algorithm, explain it. "Is there an any issue"—if , yes, how it is overcome. (2+2+1)=5 Marks

- b) Explain connection *termination mechanism* in TCP using 3-way hand shaking method.
c) Define following QoS parameter ---i) Reliability and ii) Delay

- Q-8 a) Analysed the different Closed-loop method for congestion control. (2+2+1)=5 Marks

- b) In a network, 100 stations on a pure ALOHA network share a 1Mbps channel. If frames are 1000 bits long, find *throughput* if each station is sending 10 frames per second.

- c) Define responsibilities of Physical and Session layer.

- Q-9 a) How DNS protocol works in our internet scenario. (2+2)=4 Marks
b) Explain symmetric key and asymmetric key with the help of suitable example.

- Q-10 Short Note (*any 3*)

- a) SCTP
b) Classless Addressing.
c) Packet Switching.
d) FTP
e) SNMP
f) TDMA

(3×2)= 6 marks

Roll No 1 2 3 1 2 0 2 1



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(An Autonomous Institute of National Importance under Act of Parliament)

Email: sonepatiit@gmail.com, website: www.iiitsonepat.ac.in

Probability Statistical and Numerical Methods (UCS 012)

(B.Tech. End Semester Examination, Date: May 16, 2025)

Timing: 2:00 PM to 5:00 PM IT, 2nd year (4th semester) Max mark: 50

(All questions are compulsory)

1. The velocity V of a liquid is known to vary with temperature according to a quadratic law $V = a + bT + cT^2$, find the best values of a, b and c for the following data:

T	1	2	3	4	5	6	7
V	2.31	2.01	3.80	1.66	1.55	1.47	1.41

2. (a) Define the order(rate) of convergence of Numerical Method. What is the rate of convergence of Newton Raphson Method.

- (b) The current i in electric circuit given by $i = 10e^{-t} \sin 2\pi t$ where t in seconds. Using Newton Raphson Method, find the value of t correct to 3 decimal places for $i = 2$ amp.

3. Using the Gauss's backward interpolation formula to obtain y_{28} ; given that $y_{20} = 49225$, $y_{25} = 48316$, $y_{30} = 47236$, $y_{35} = 45926$, $y_{40} = 44306$

4. The distance covered by an athlete for the 50 metre race is given in the following table: Determine the speed and acceleration of athlete at $t = 5$ second.

Time(sec)	0	1	2	3	4	5	6
Distance in (metre)	0	2.5	8.5	15.5	24.5	36.5	50

5. (a) When do we need numerical integration?

- (b) A rocket is launched from the ground vertically upwards. Its acceleration (a) is registered the first 80 seconds and is given in the table. Estimate the velocity and the height of the rocket at time $t = 80$ seconds. Use trapezoidal rule as well as Simpson's 1/3 rule.

Time(s)	0	10	20	30	40	50	60	70	80
$a(m/s^2)$	30.0	31.00	33.44	35.47	37.75	40.33	43.29	46.69	50.47

0 1 2 3 4 5 6 7 8

Handwritten notes and calculations:

$$v = \frac{ds}{dt}$$

$$v = \frac{ds}{dt}$$

$$v = \frac{ds}{dt}$$

$$v = \frac{ds}{dt}$$

6. Given $\frac{dy}{dx} = x^3 + y$, $y(0) = 2$, compute $y(0.2)$, $y(0.4)$, $y(0.6)$ using the Runge-Kutta method of fourth order. Compare with the exact solution.

7. (a) VLSI chips, essential to the running of a computer system, fail in accordance with a Poisson distribution with the rate of one chip in about 5 weeks. If there are two spare chips on hand, and if a new supply will arrive in 8 weeks. What is the probability that during the next 8 weeks the system will be down for a week or more, owing to a lack of chips?

(b) If X is a poisson variate such that $P(x=2) = 45.P(x=6) - 3.P(x=4)$. Find (i) $P(x \geq 1)$ (ii) $P(x < 2)$

8. A company finds that the time taken by one of its engineers to complete or repair job has a normal distribution with mean 40 minutes and standard deviation 5 minutes. State what proportion of jobs take: (a) less than 35 minutes (b) more than 48 minutes. The company charges Rs. 20 if the job takes less than 35 minutes, Rs. 40 if it takes between 35 and 48 minutes and Rs. 70 if it takes more than 48 minutes. Find the average charge for a repair job.

9. Write a short note on the following

- i) Level of significance, left tailed test, right tailed test, two tailed test
- ii) Critical region
- iii) Type I and type II errors
- iv) Test of significance for difference of mean of two large samples
- v) Null Hypothesis and Alternative Hypothesis

10. A company has the head office at Calcutta and a branch at Bombay. The personnel director wanted to know if the workers at the two places would like the introduction of a new, plan of work and a survey was conducted for this purpose. Out of a sample of 500 workers at Calcutta, 62% favoured the new plan. At Bombay out of a sample of 400 workers, 41% were against the new plan. Is there any significant difference between the two groups in their attitude towards the new plan.

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Email: sonenpatiiit@gmail.com, website: www.iiitsonepat.ac.in
B.Tech. End Semester Examination, May'2025.

Subject: Computer Networks

Subject Code: ITC404

Roll No.

1	2	3	1	2	0	2	1
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Branch: IT

Max. Marks: 50

Time- 3 Hours

Note: All questions are compulsory, and your answer should include a rough sketch, if necessary.

Q-1

5+3+2= 10

- (a) Explain the ISO-OSI model of computer networks with a neat diagram
- (b) Distinguish between point-to-point links and multi-point communication. Give relevant diagrams
- (c) "QAM modulation technique provides the highest spectral efficiency"— Justify your answer.

Q-2

5+3+2= 10

- a) Describe and compare ALOHA, Slotted ALOHA, CSMA, CSMA/CD, and CSMA/CA. Mention where each protocol is used

b) What are the window sizes of Go-Back-N ARQ and Selective Repeated. Which is the more efficient, selective repeat or Go-Back-N ARQ? Justify your answer.

c) Explain the difference between a one's complement and two's complement checksum.

Q-3

5+3+2= 10

• a) IIIT Sonepat is granted an address block from ISP, starting with 150.80.0.0/16. IIIT Sonepat wants to distribute these blocks to 2600 users as follows.

1. The first group has 200 for faculty of the IIIT Sonepat; each needs 128 addresses.
2. The second group has 400 for Office of IIIT Sonepat; each needs 16 addresses.
3. The third group has 2000 for students of IIIT Sonepat; Each needs 4 addresses.

Design the subnet and give the slash notation for each subblock. Find out how many addresses are still available after these allocations.

b) Compare IPv4 and IPv6 in terms of address structure, header format, and features.

c) Describe with a diagram TCP connection termination scenario

- Q-4 a) e) Short note (any 2)
i) E-mail protocol stack
~~ii) Public and Private Key~~
~~iii) Line coding~~
iv) IGMP

$$5+3+2=10$$

b) Compare FDM and TDM. Explain the working principle of the PCM technique.

c) How is dialog control managed at the session layer? What is CIDR ?

- Q-5 a) A sender is using Go-Back-N with a window size of 4 and sequence numbers modulo 8. What is the range of sequence numbers that can be sent before an acknowledgment is received?

$$4 \times 2^{\frac{1}{2}} = 10$$

b) A TCP connection starts in slow start. The congestion window is initially 1 MSS and doubles every RTT. How many RTTs are needed before the window reaches 64 MSS? Total segments sent until that point?

c) TCP requires a three-way handshake. If RTT is 150 ms and one MSS is sent after handshake, what is the total delay to send 1 KB of data?

d) Measure the performance of a telephone line (4-KHz of bandwidth). When the signal is 20V, the noise is 6 mV. What is the maximum data rate supported by this telephone line?

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77 sec

(Total pages: 02)

Roll No.	1	2	3	1	2	0	2	1
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Email: sonapatiit@gmail.com, website: www.iiitsonapat.ac.in

Department of Information Technology (IT)

Microprocessor and Interfacing (UCS015)

End Semester Examination (May 23, 2025)

Time: 02:00 PM to 05:00 PM Semester-IV (Regular) Total marks: 50

Note: All questions are compulsory

1. (a) List the timing and control signals of 8085 Microprocessor and explain briefly. (2)
(b) Draw the architecture of 8085 Microprocessor and explain the function of each block, briefly. (3)
2. (a) What is a buffer? Why is it needed in microprocessor system? (2)
(b) How the address and data lines are demultiplexed in 8085 microprocessor? Explain it with diagram. (3)
3. (a) What is the difference between Subtraction and compare instruction of 8085 microprocessor? (2)
(b) Define an addressing mode. What are the addressing modes available in 8085 microprocessor? Give an example for each. (3)
4. (a) A software delay subroutine is written as given below (2)

```
DELAY: MVI H, FFH
        MVI L, FFH
LOOP:   DCR L
        JNZ LOOP
        DCR H
        JNZ LOOP
```

How many times DCR L instruction will be executed? Explain it properly.

- (b) Explain the following instructions with an example: (3)
 - XCHG, DAA, RAL, DI, XRA M, PCHL

T state in
XTH2

5. Define an interrupt. What is the difference between software and hardware interrupts? Differentiate between all hardware interrupts of 8085 microprocessor. (5)
6. (a) Frame the command words *ICW1*, *ICW2*, and *OCW1* for initializing single 8259 with call address interval of 8 and for level triggered interrupt. Also, unmask all the interrupt inputs. The desired vector address is 2060H. (2)
- (b) Explain the functional block diagram of IC 8259- Programmable Interrupt Controller in details. (3)
7. (a) What are the two control words needed to initialize IC 8255- Programmable Peripheral Interface. Explain it. (2)
- (b) Explain the internal block diagram of IC 8255- Programmable Peripheral Interface in details. (3)
8. (a) List the differences between 8085 and 8086 microprocessors. (2)
- (b) Explain the architecture of 8086 microprocessor with proper diagram. (3)
9. Write an assembly language program to multiply two numbers of 8-bit data stored in memory locations 2200H and 2201H with flow-chart. Store the product at memory locations 2202H and 2203H. (5)
10. (a) The crystal frequency of 8085 microprocessor is 6 MHz. What is the time required to execute instruction XTHL over this microprocessor? (2)
- (b) An 8085 assembly language program is given below (3)
- Line 1: MVI A, B5H
Line 2: MVI B, 0EH ,
Line 3: XRI 69H ,
Line 4: ADD B
Line 5: ANI 9BH
Line 6: CPI 9FH
Line 7: STA 3010H
Line 8: HLT
- What will be the contents of the accumulator just after execution of the ADD instruction in line 4?
 - How many Bytes and T-states are required to execute the code?.

End of question paper



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Email: sonepatiiit@gmail.com website: <https://iiitsonepat.ac.in>

End Semester Examination

Branch- IT, 2nd Year (4th Semester)

Subject: Compiler Design (UIT406)

Maximum Marks- 50 marks

Note: All questions are compulsory.

Time: 3 hours

Roll No. 12312021

- Q.1(a) Differentiate between compiler and interpreter. Explain the need for dividing the compilation process [5]
into various phases. Explain the various phases of a compiler with help of the following program.

```
main(){  
    int a, b;  
    float c, d;  
    c=a+b*10; d=a-b/2.0;  
}
```

- Q.1(b) Consider the grammar: $E \rightarrow TE'$, $E' \rightarrow E \mid e$, $T \rightarrow FT'$, $T' \rightarrow T \mid e$, $F \rightarrow PF'$, $F' \rightarrow *F' \mid e$, $P \rightarrow (E) \mid a \mid b \mid e$ [5]
Here, e stands for Null (epsilon).
(i) Compute FIRST & FOLLOW for each non terminal of the above grammar.
(ii) Find the predictive parsing table and conclude whether the grammar is LL (1) or not.

- Q.2(a) What are steps to eliminate left recursion and left factoring? Remove left factoring and left recursion [5]
of the following grammar:

(i) $S \rightarrow aAd \mid aB$ (ii) $A \rightarrow Br \mid x$
 $A \rightarrow a \mid ab$ $B \rightarrow Cs \mid y$
 $B \rightarrow ccd \mid ddc$ $C \rightarrow At \mid z$

- Q.2(b) Verify the string $id + id * id$ is member of the following grammar or not, by using LL(1) algorithm: [5]

$E \rightarrow TX$
 $X \rightarrow +TX \mid \epsilon$
 $T \rightarrow FY$
 $Y \rightarrow *FY \mid \epsilon$
 $F \rightarrow id$

- Q.3(a) Construct an equivalent grammar after removing left recursion: $E \rightarrow Ea \mid Eb \mid b \mid a$ [5]
Explain shift-reduce and reduce-reduce conflicts in SLR(1) with an example.

- Q.3(b) Construct LALR(1) table and verify the following grammar is LALR(1) or not: [5]

$S \rightarrow Aa$
 $S \rightarrow bAc$
 $S \rightarrow Bc$
 $S \rightarrow bBa$
 $A \rightarrow d$
 $B \rightarrow d$

- Q.4(a) Construct operator precedence parsing table for the following grammar [5]

$A \rightarrow B * A \mid B$
 $B \rightarrow B + C \mid C$
 $C \rightarrow D @ C \mid D$
 $D \rightarrow d$

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Q.4(b) Differentiate between LL(k) and LR(k) parsers, where k denotes the number of look-ahead tokens. Can you consider k=0 for LL(k)? If so, how does the parser work? Explain why LR parsing is attractive. [5]

Q.5(a) Consider the grammar
 $S \rightarrow Aa | bAc | Bc | bBa$
 $A \rightarrow d$
 $B \rightarrow d$
Show the grammar is LR (1) but not LALR (1). [5]

Q.5(b) Given a Grammar
 $S \rightarrow L = R$
 $S \rightarrow R$
 $L \rightarrow *R$
 $L \rightarrow id$
 $R \rightarrow L$

Construct LR(1) set of items for this Grammar and Also construct the Canonical LR Parser for it.